AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 32 · STUDENT LESSON

Build a Paper Prototype

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Make the cards. Run them literally. Keep the actual trace.

Build the activity prototype, separate expectations from observations and diagnose the first mismatch without rewriting history.

Keep one habit: Save the version → predict independently → record the actual run → label the repair.

MAIN LESSON · SECTION 1

Can someone else run your written process?

> Turn the activity flow into instructions, visible state and an honest run record.

Start with the hidden human repair

An operator who silently fixes missing instructions can make a weak prototype look successful. Save the written version, predict the required result independently and record what actually happens.

Recall when a failed candidate permits no match

When one candidate fails, what must happen next, and when is no match justified?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can build a runnable paper **prototype** with explicit **instruction**s and visible state.
- I can record what actually happens without silently repairing the **instruction**s.
- I can use a **trace** to distinguish an **instruction** defect from an execution mistake.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

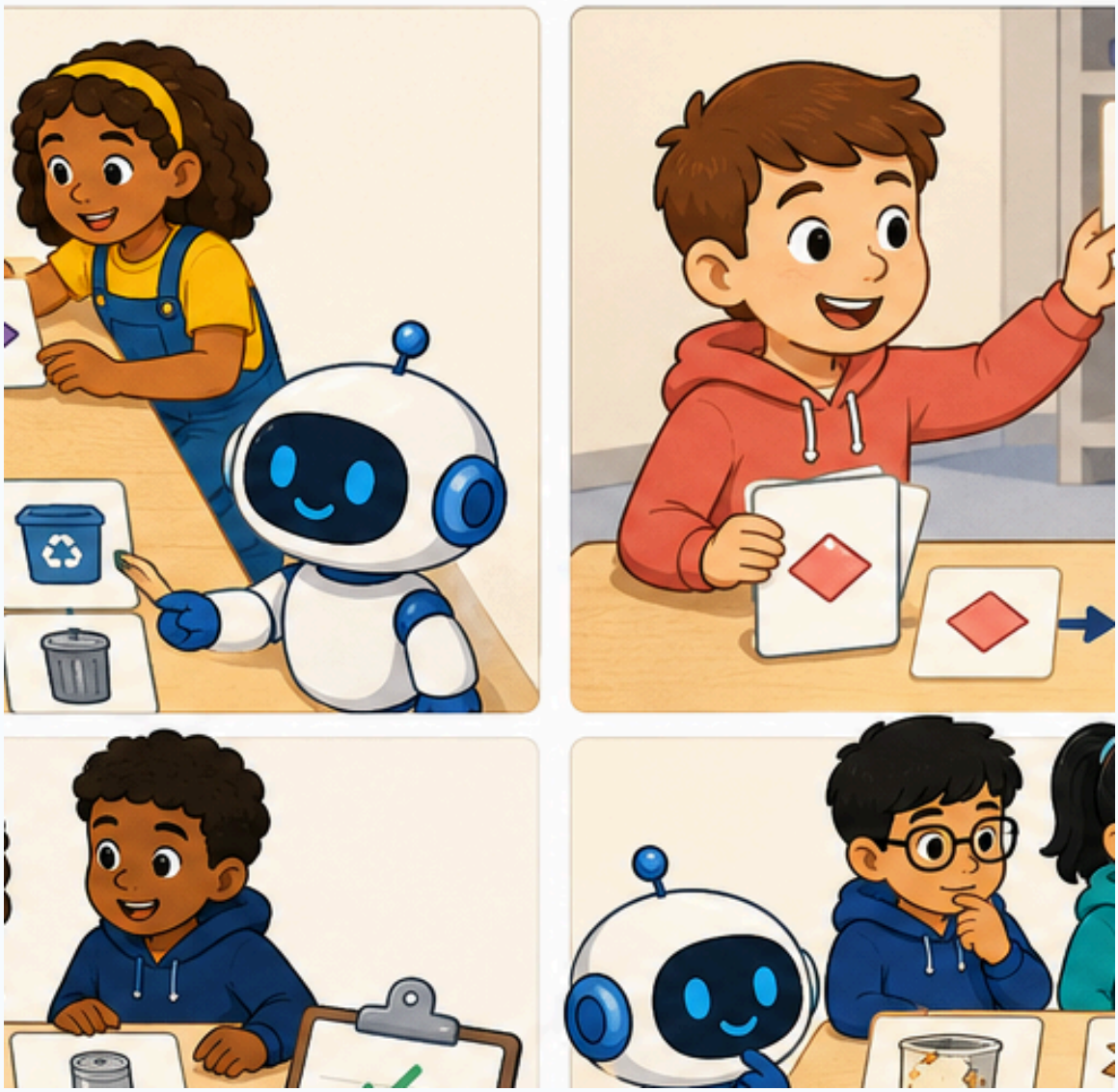
Remember before reading

Without opening the old lesson, explain one key idea from Design the Decision **Process**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompts Are Design **Instruction**s. Give an example and a mistake that the idea helps you avoid.

A **prototype** tests the idea before polish or technology.
</details>



LEARNING TARGET Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes.

What does this **prototype** demonstrate?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can build and operate a paper **prototype** while recording actual **trace**s.

MAIN LESSON · SECTION 2

Build the activity prototype from the same contract

> Continue the paper project with catalogue B version 1 and labelled instructions.

Define an executable paper prototype

This week you will turn the decision **process** into a paper **prototype**. One partner operates the **instruction**s and another records the steps.

A working **prototype** needs more than a polished drawing: someone else must be able to follow it without asking its author to supply missing rules. Keep the version you ran so the **trace** can be checked against it.

Copy the four catalogue cards

Copy this catalogue in order onto four cards: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These are fictional paper-model values with materials assumed available. Label the catalogue B version 1.

Preserve the full input/output contract

Keep the Week 30 contract: **input** is a whole number of available minutes from 0 to 30, and Quiet required Yes or No.

An activity fits when its time is no more than the available minutes AND either quiet is not required or the activity is quiet. No means either kind is allowed. Suggest the first fit with a reason; otherwise report no match.

Missing or invalid **input** needs clarification. Suggestions do not book materials or publish anything.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY One student presents an item card while another follows the written sorter **process** exactly and returns the documented **output**.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Prepare cards, a marker and a trace sheet

> The operator uses only the saved instructions, with state reset for every request.

Inspect the prototype’s three stages

A paper run that can be checked

PREPARE THE RUN	OPERATE ONE STEP	KEEP THE EVIDENCE
Request and version labels	Validate the two fields	Record pass or clarification
Marker at current row	Check time AND quiet	Record checks and next step
First fit or end of list	Suggest or report no match	Record output and stop

The request stays fixed while the marker advances. Keep the instruction version and trace together.

Make the materials and record fields

Make a request card with Available minutes and Quiet required fields, a marker for the current catalogue row, and a **trace** sheet. Head the sheet Run ID, **prototype** version, catalogue version and request.

Below it use columns Step, current row, time check, quiet check and next action or **output**. Use not checked when validation stops the search.

Write P1 version 1's instruction cards

Write **instruction** cards P1 version 1. Card 1: validate both fields; if invalid, name the field needing correction and stop; otherwise put the marker at B1.

Card 2: check the marked row against both compatibility conditions; record both checks. If both pass, suggest that row with the reason and stop.

Card 3: if either fails and another row remains, advance exactly one row and return to Card 2; if no row remains, report no match and stop. Begin each new request at Card 1 with a cleared **output**.

Explain what the operator follows

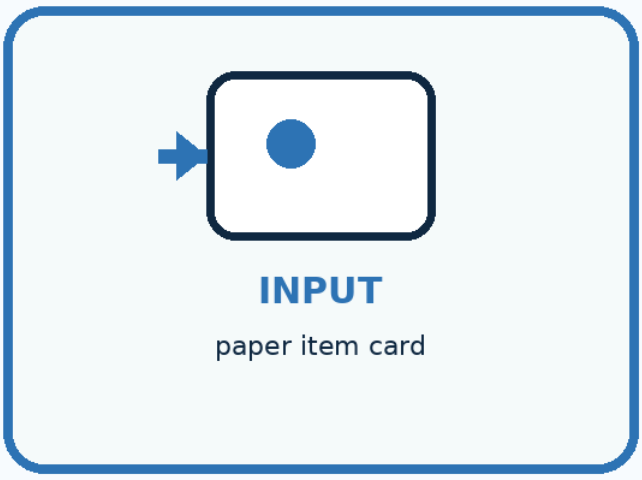
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

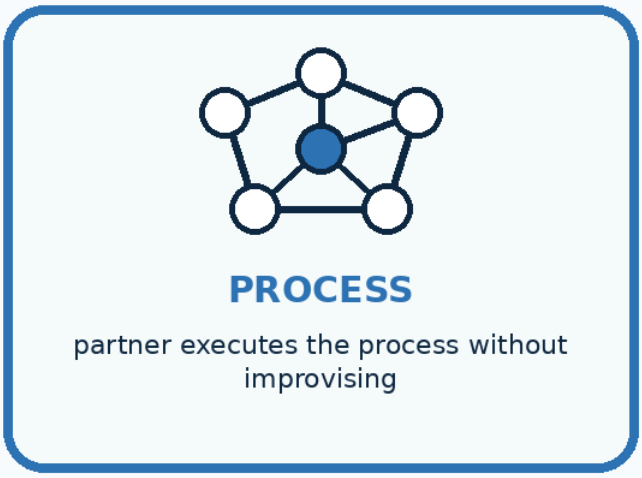
▼ **Earlier wording, alternatives and picture transcripts**

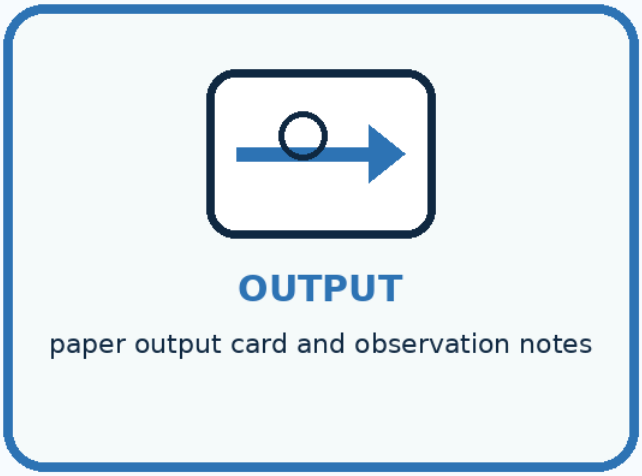
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT paper item card

PROCESS partner executes the **process** without improvising

OUTPUT paper **output** card and **observation** notes

CORE IDEA A **prototype** tests the idea before polish or technology.

UNDER THE HOOD A low-fidelity **prototype** externalizes assumptions cheaply.

Wizard-of-Oz **simulation** is valid when the human **operator** follows the specified **process** and does not secretly add judgment unavailable to the proposed system.

Draw a fresh example with all three stages:

A closer explanation

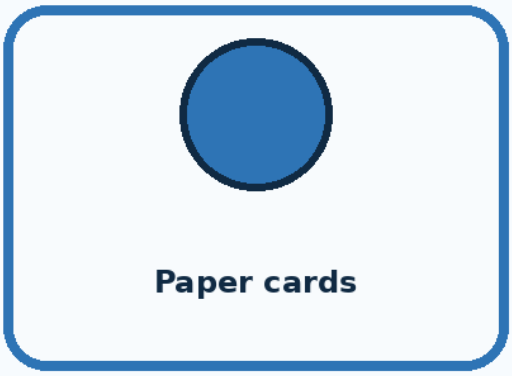
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare a literal paper run with hidden improvisation

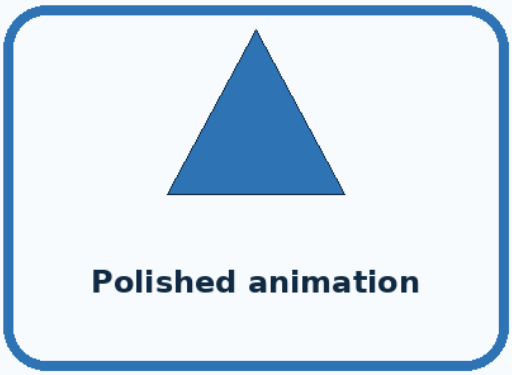
> Polish and a helpful operator do not establish a complete written process.

Pictures paired with their evidence



The case to investigate

CASE 1 — Paper cards Clue: fast to revise. Meaning: good for logic test.



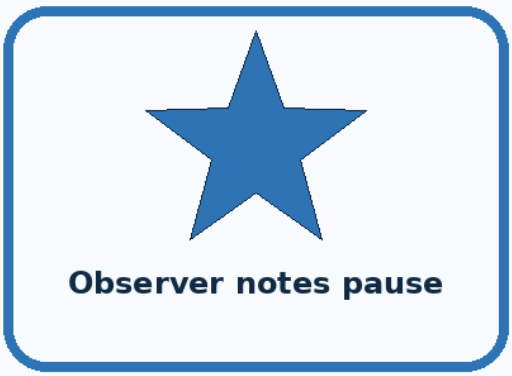
The case to investigate

CASE 2 — Polished animation Clue: looks complete. Meaning: may hide weak **process**.



The case to investigate

CASE 3 — **Operator** improvises Clue: adds secret knowledge. Meaning: invalid **simulation**.



The case to investigate

CASE 4 — Observer notes pause Clue: reveals confusion. Meaning: useful **evidence**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

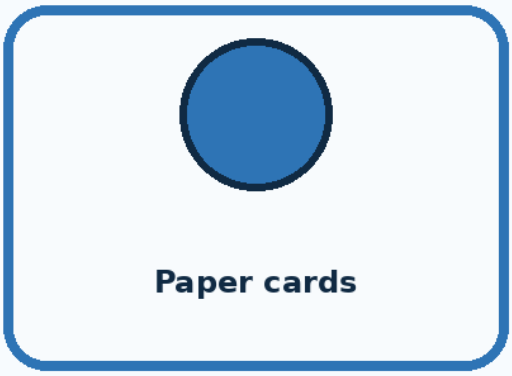
BOUNDARY A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

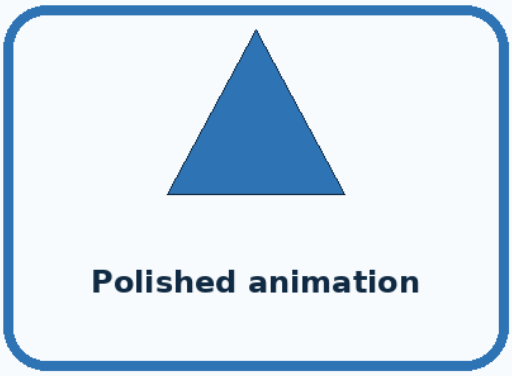
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

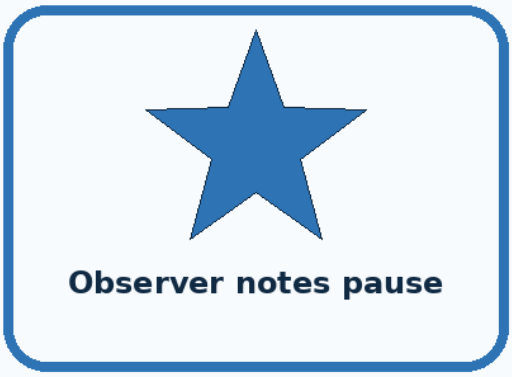
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Paper cards Clue: fast to revise. Meaning: good for logic test.

CASE 2 — Polished animation Clue: looks complete. Meaning: may hide weak **process**.

CASE 3 — **Operator** improvises Clue: adds secret knowledge. Meaning: invalid **simulation**.

CASE 4 — Observer notes pause Clue: reveals confusion. Meaning: useful **evidence**.

Change one thing in the pictured case

Change: Insert “if capacity is below count, continue” before accepting a bin.

Prediction: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

Trace the separate X/Y bin prototype defect

> Distinguish the required baseline result from the actual faulty-card result.

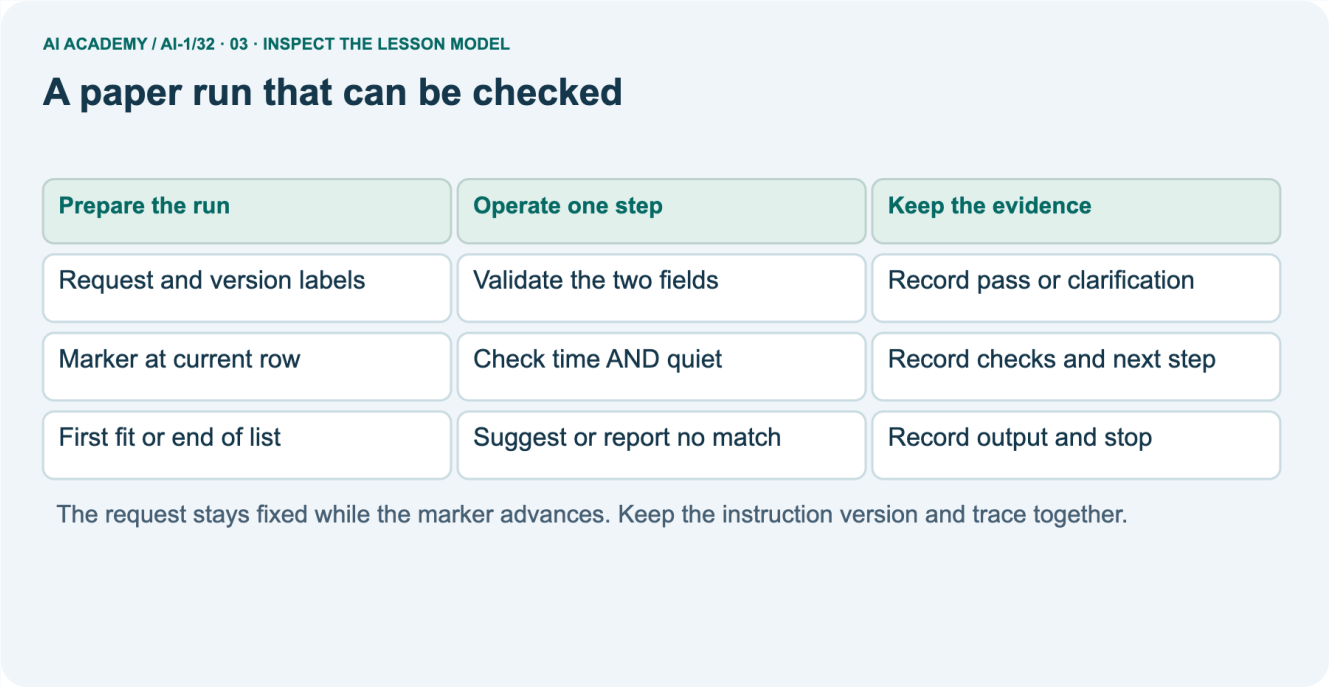
Read the bin contract and faulty instructions

A fictional team turns a storage-bin specification into movable **instruction** cards.

Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

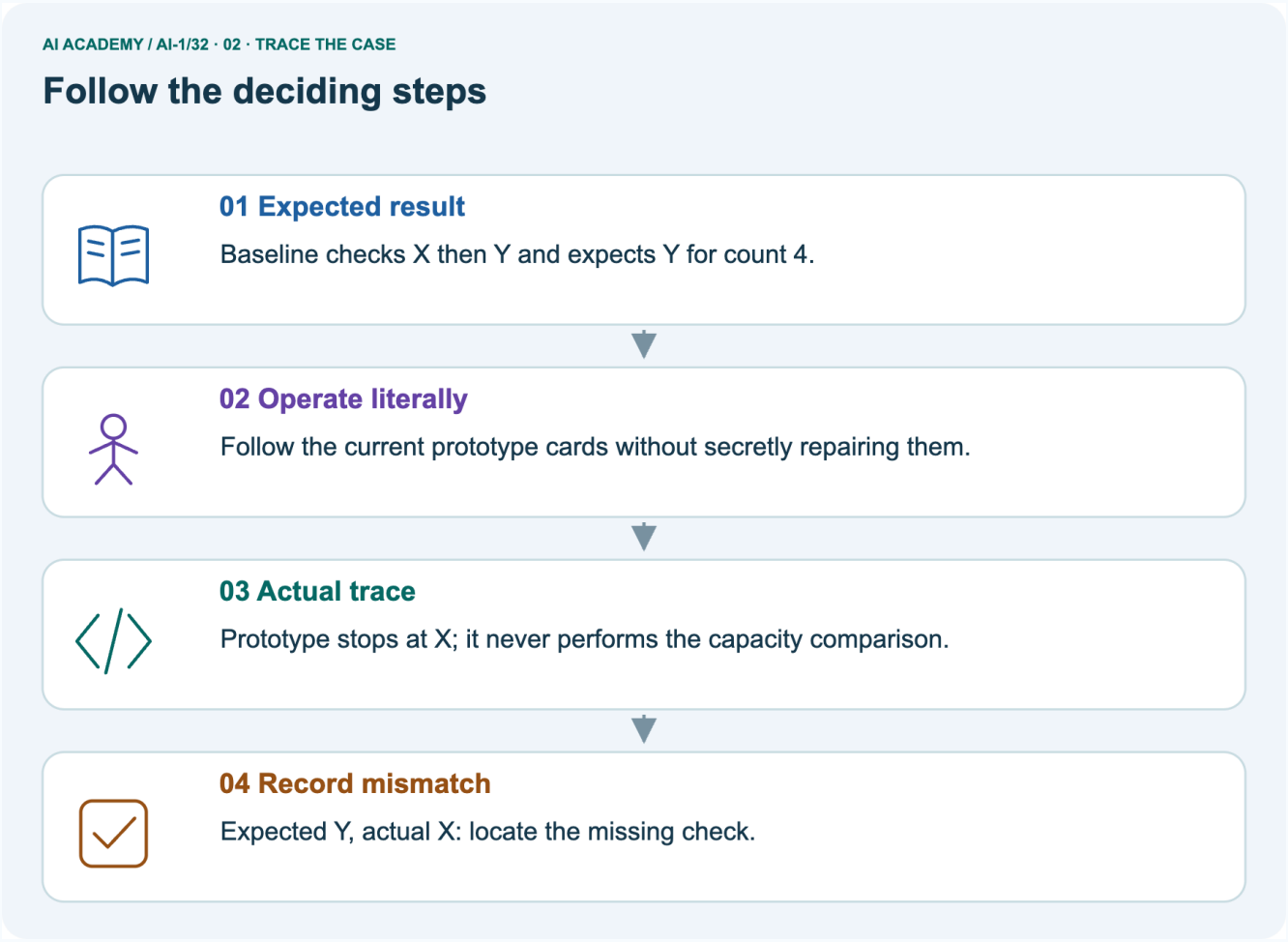
For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

Return to the activity prototype's trace structure



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the actual mismatch without repairing it silently



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Expected result	Baseline checks X then Y and expects Y for count 4.
2. Operate literally	Follow the current prototype cards without secretly repairing them.
3. Actual trace	Prototype stops at X; it never performs the capacity comparison.
4. Record mismatch	Expected Y, actual X: locate the missing check.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain what the before-and-after log must preserve

- A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

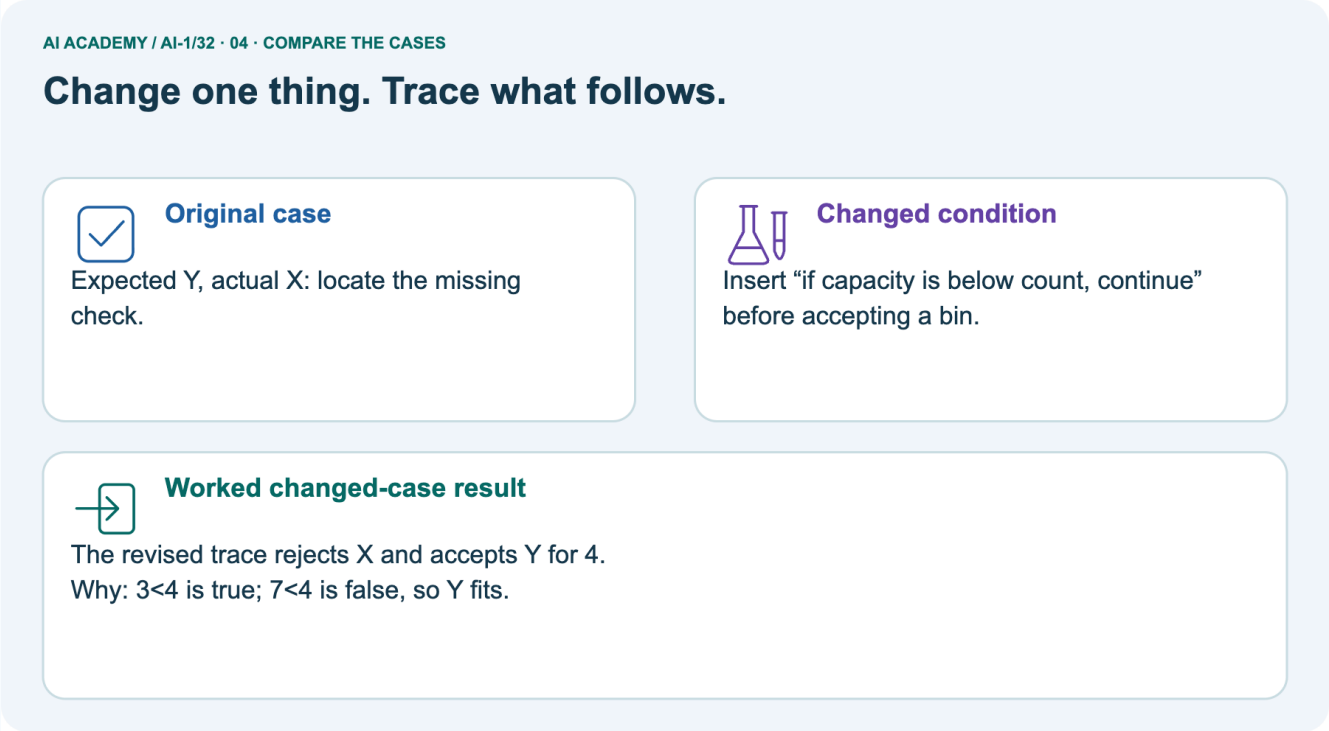
  Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

- Read the labelled picture: What should appear in the test log before and after the repair?

Predict the explicit capacity-check repair

- Before opening the repaired-bin picture, add a capacity comparison before accepting a bin and continue to the next row when capacity is too small. Predict the new trace for count 4. What must remain saved from the original run?

▼ Check the repaired trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit what one repair establishes

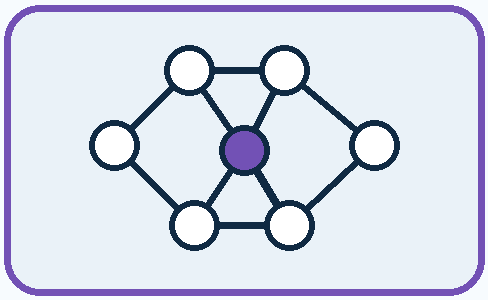
One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE One student presents an item card while another follows the written sorter **process** exactly and returns the documented **output**.

STEP 1 — OBSERVE paper item card

STEP 2 — **TRACE** partner executes the **process** without improvising

STEP 3 — CONCLUDE paper **output** card and **observation** notes

MODEL REASONING The **operator** must use only documented **input**s and steps. Any question, pause, or improvisation reveals a missing rule, unclear label, or unsupported **input**. Revise the paper flow before adding technology.

DO NOT OVERCLAIM A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

What new **evidence** could change this answer?

A closer explanation

Before operating P1, use the full manual checklist to write the **expected result** for 9 minutes, Quiet required No. Then run the cards literally. Do not replace a surprising result with the expected one.

Recording disagreement gives you something to investigate; hiding it removes useful **evidence**.

Trace step for 9 minutes No	Checks recorded	Next action or output
1 Validate	Whole minutes in range; No accepted	Place marker at B1
2 Check B1	10 is greater than 9; quiet unrestricted	Move to B2
3 Check B2	8 is no more than 9; quiet unrestricted	Suggest B2 with reason; stop

This **trace** agrees with the expected B2 result. It shows that this run followed the **instruction**s. One matching result does not prove that every request will work or that the **prototype** is faster than the manual checklist.

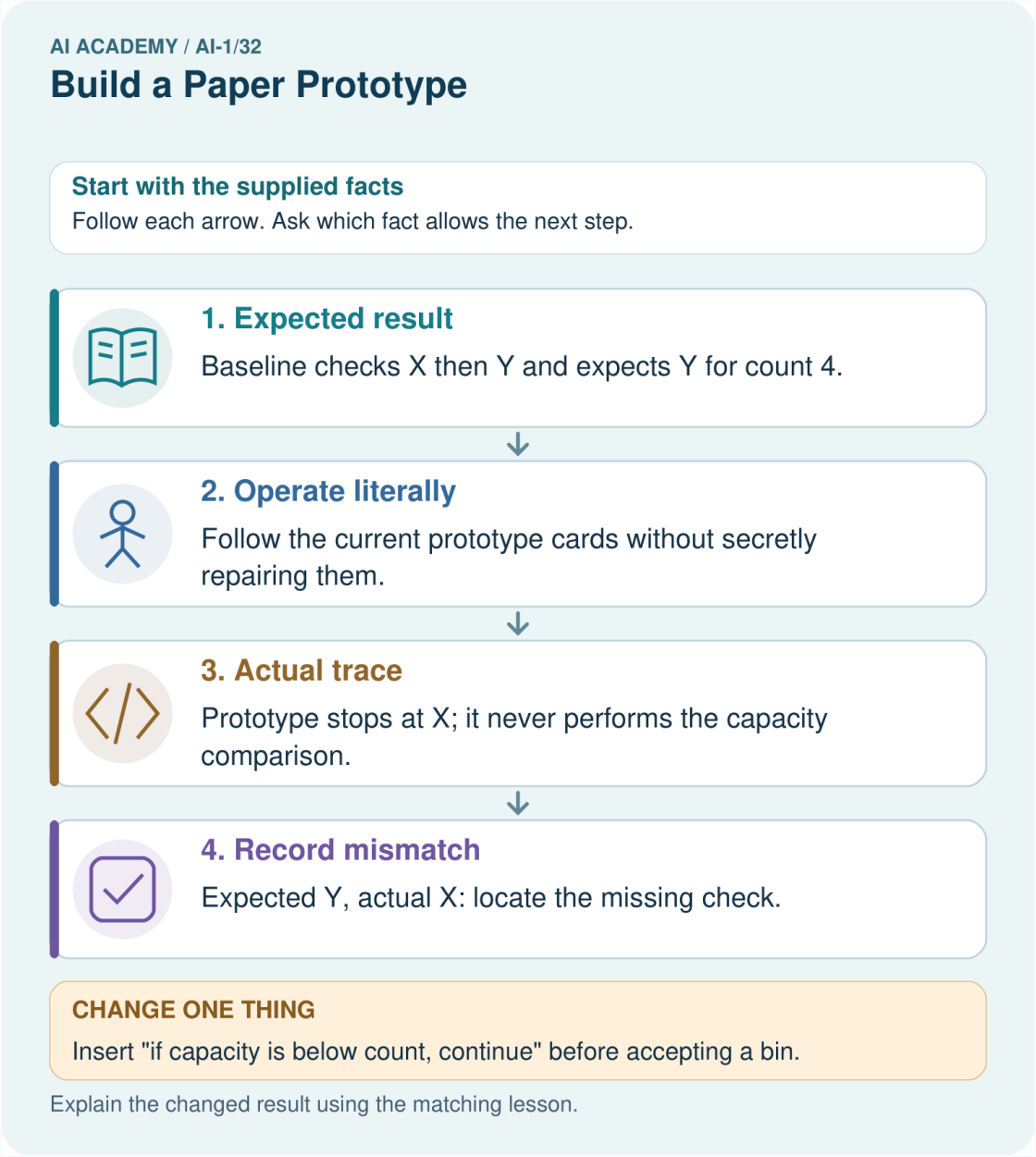
This is a system of **written rule**s, not a model trained on examples.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Build a Paper Prototype

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Insert “if capacity is below count, continue” before accepting a bin.
Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Hand over the prototype without extra hints

> Save the version and every run so another person can inspect the result.

Swap roles and record ambiguity

Swap roles and give your partner the request card, catalogue, **instruction** cards and blank **trace** sheet. Let them operate without extra hints from the author.

Record any point where they cannot decide what to do from the written **instruction**s. Label and save changes, then repeat the affected run.

Keep development cases separate from any cases your teacher has reserved for a later evaluation.

Keep a repeatable handover and all disagreements

Your handover should let someone repeat a run using the same **input**s and versions and check why it stopped. Keep all results, including disagreements. Next week you will organise several routine cases into a test record.

A neat interface is useful, but correctness depends on validation, both conditions, the first-match rule and a clear ending.

Repair both tempting claims

- CLAIM A A **prototype** must look like the final product.
- CLAIM B If a person can perform the task, the written **process** is complete.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Paper cards	fast to revise	good for logic test
Polished animation	looks complete	may hide weak process
Operator improvises	adds secret knowledge	invalid simulation
Observer notes pause	reveals confusion	useful evidence

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **A saved trace says “P1 version 1; 11 minutes No; started at B2; suggested B2.” Explain the error using the written cards. State what evidence to keep and what should happen in a new correct run.**

Practise: distinguish a bad instruction from a bad execution

> Inspect the saved version before diagnosing the first mismatch.

Audit the marker-reset mistake

Suppose the **operator** starts a fresh 11-minute No request at B2 because the marker was left there. They return B2. The required first fit is B1, and Card 1 explicitly says to restart at B1.

The **trace** therefore reveals an execution mistake. If the written **instruction**s had omitted the reset, that would instead be an **instruction** defect. Inspect the saved version before deciding.

Choose the correct repair and preserve the first trace

Preserve the first **trace**. For an execution mistake, rerun the same written version correctly and label the new run. For an **instruction** defect, save a corrected version with a new label, state what changed, and rerun.

Never rewrite an old **observed result** to make a version appear successful.

Run and record the first match in W2

Use workbook W2 for the connected example “Building and tracing a paper **prototype**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the answer-is-enough claim

- Use the worked case for Building and tracing a paper **prototype**. Proposed explanation: “A correct-looking answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the before-and-after bin log

What should appear in the test log before and after the repair?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A **prototype** tests the idea before polish or technology.

Use a fresh example to explain **prototype**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: If a person can perform the task, the written **process** is complete.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **Run P1 for 8 minutes with Quiet required Yes. Record every row, both checks and the final output. Then consider a blank Quiet required field: which card stops that run and how many candidates should be checked?**

Practise with less help: Run and record the first match

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate a literal prototype handover

> Use complete materials and keep expected outputs separate from observations.

Prepare the shared materials and requests

Partner-as-computer paper run using catalogue B version 1 and the P1 version 1 instruction cards supplied in this lesson. Prepare request cards (9 minutes,No), (11 minutes,No), (7 minutes,No), and (blank,Yes). Before operating, use the full manual baseline to write each expected output on a separate sheet; do not show it to the operator while they run.

Predict, operate, record and diagnose

1. Assign reader, operator and recorder roles, or combine reader/recorder in a pair. Give the operator only the request, catalogue, saved cards and marker. Record the actual validation, row checks, next steps and output, including pauses or inability to proceed. Reset marker/output for each new request. Compare with expectations afterwards; inspect the saved cards to distinguish an instruction defect from an execution mistake. Keep the first trace, label a rerun and any revised version, and swap roles. Do not invent an observed run if you have only planned it.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Retain every run and version

Case	Prediction	Observed result	What it means
Normal			
Edge			

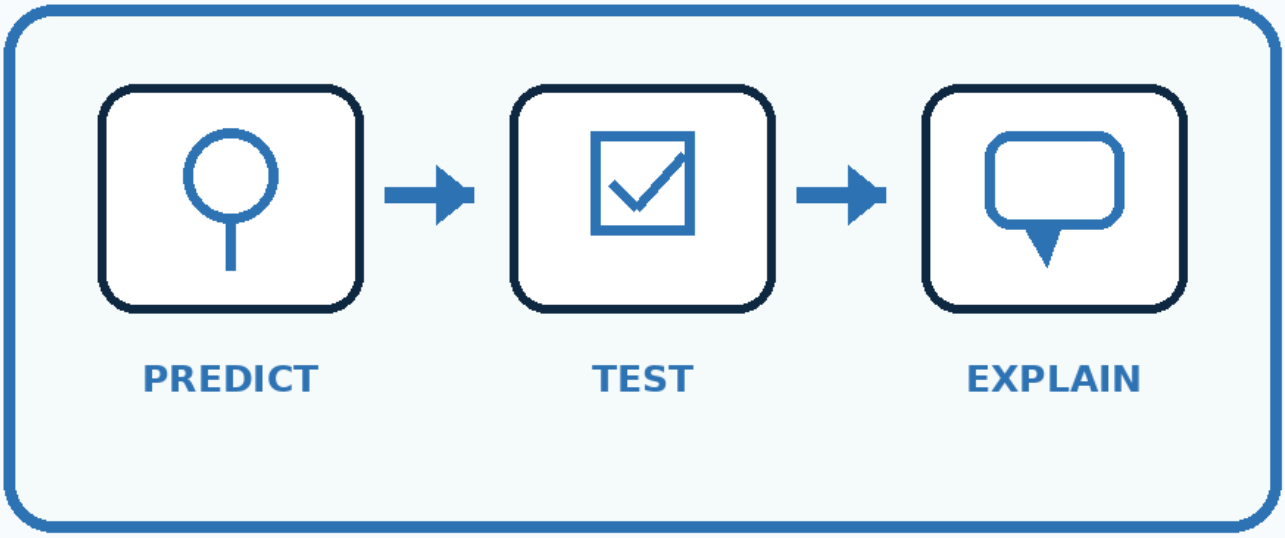
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Partner-as-Computer Prototype

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Run the sorter with a partner acting as the computer.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: make and trace the equipment prototype

> Use E1/E2 and preserve separate expected and actual run records.

The independent catalogue and run tasks

Independently make and **trace instruction** cards for this paper equipment catalogue: E1 first, capacity 3 people; E2 second, capacity 6. Valid **input** is a whole group size from 1 to 8.

Suggest the first item whose capacity is at least the group size; no booking occurs. Include invalid **input** and no match. Write **expected result**s, then operate your cards for groups of 5 and 7.

Record both runs and explain every stopping point.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build a new route through a small catalogue

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Your paper **prototype** shows an **input** card, a decision card and an **output** card. Explain what a person must simulate and what the **prototype** cannot prove.

My answer, with a **trace**, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and **expected result**

> **Write or draw your reasoning and evidence.**

Connect to your project

Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Assign three people as **input** reader, rule follower and **output** recorder. Run the repaired sorter on paper, metal and an unreadable description. Record expected and observed outcomes. What does this **prototype** demonstrate?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Transfer the prototype record to the sorter portfolio

> Keep each context’s materials, versions and runs distinct.

Connect the two contexts

The connected final sequence builds the activity prototype. This section transfers the same materials-and-trace method to your existing sorter portfolio, without requiring a second implementation. Keep catalogue, instructions and observed traces labelled by project and version; an activity-run result is not a sorter-run result.

This week’s sorter materials

THIS WEEK'S CHECKPOINT Build the complete paper sorter: **input** cards, **process** guide, **output** cards, not-sure card, and test log.

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s.

Inspect inputs, execution and uncertainty

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Prevent hidden operator knowledge

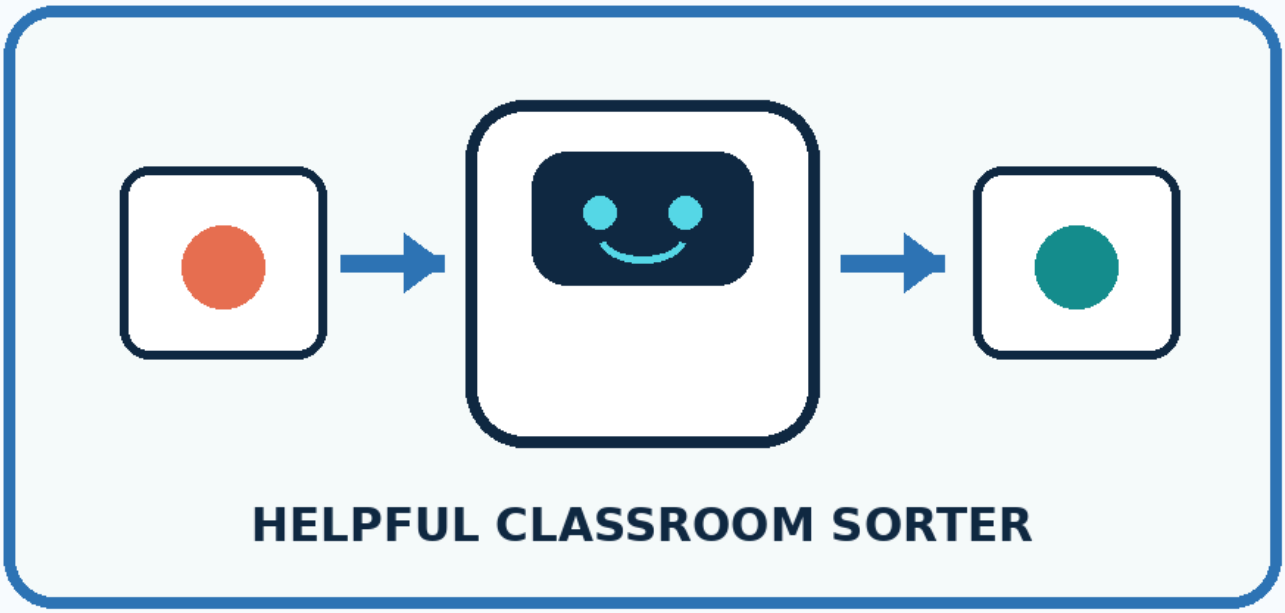
RESPONSIBLE DESIGN A helpful **operator** can accidentally make a weak design look successful by using unstated human knowledge.

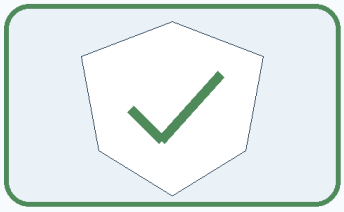
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain what your paper run demonstrates

> Close your notes and distinguish intended behaviour, actual steps and a limited conclusion.

Your exit explanation

- Close your notes. Explain this goal in your own words: Build a runnable paper **prototype** with explicit **instruction**s and visible state

My deciding step, evidence and one remaining question

Choose the next changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
prototype	early representation used to test an idea	
low fidelity	simple version focused on function	
simulation	acting out system behavior	
operator	person executing the prototype process	

Terms for the connected explanation

Word	Meaning
Instruction	A precise action or decision for the operator to follow
Trace	A record of the steps and state during one run
Expected result	The result required by the agreed specification

Word	Meaning
Observed result	What actually happened when this version was run

Repair a claim using a word

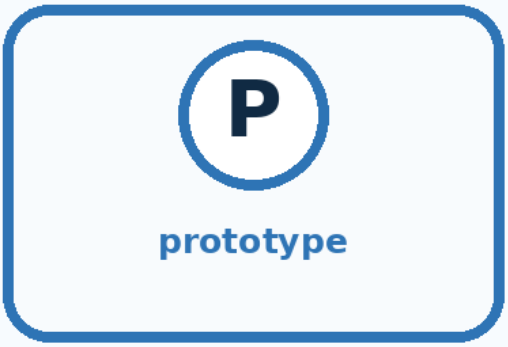
- CLAIM TO REPAIR A **prototype** must look like the final product.
- Use one vocabulary word and one piece of **evidence** to repair it:

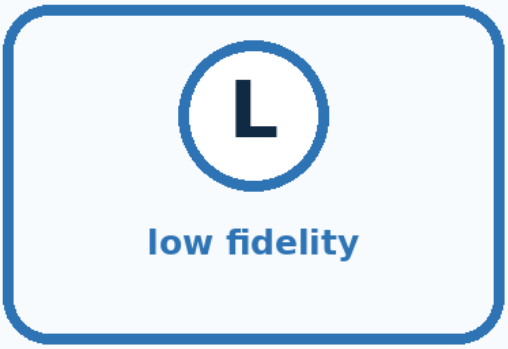
▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

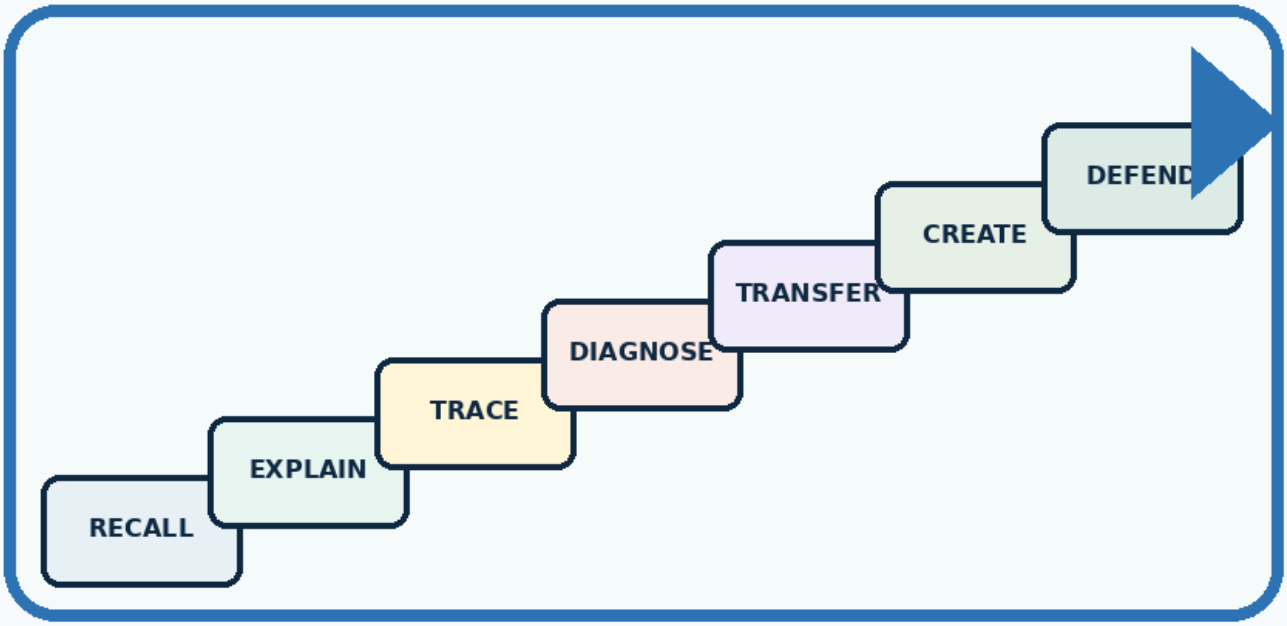
- RECALL Define **prototype**.
- EXPLAIN A **prototype** tests the idea before polish or technology.
- **TRACE** Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A **prototype** must look like the final product.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Build a new route through a small catalogue is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

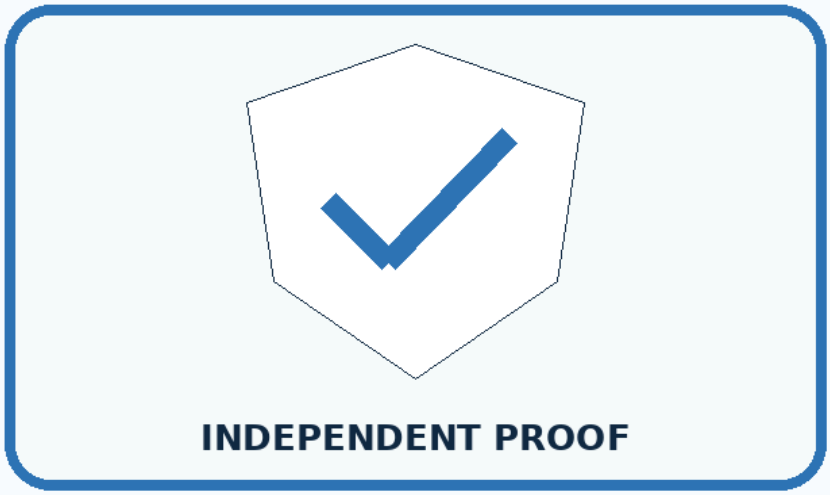
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Build a new route through a small catalogue is the selected independent case.



EXPLANATION CHECK Build runnable instruction cards, predict from the contract and preserve actual state, outputs and stopping reasons.

Independent **evidence**:

SELF-CHECK The **operator** must use only documented **input**s and steps. Any question, pause, or improvisation reveals a missing rule, unclear label, or unsupported **input**. Revise the paper flow before adding technology.

NEXT BRIDGE Carry today's **evidence** habit into: Test Normal Cases.

▼ **Extra practice: Act out a paper prototype**

REFERENCE · SECTION 13

Extra practice: Act out a paper prototype

> Try the question before opening its worked explanation.

Read the supplied case and question

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

How can a learner simulate it?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

Evidence or stage	Value or result
Input card	Observe material
Decision cards	Follow written checks
Token	Place in paper, plastic or review tray
Trace	Record each decision

▼ Earlier wording, alternatives and picture transcripts

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Act out a paper prototype

A paper **prototype** has **input** cards, decision cards and paper, plastic and review trays.

How can a learner simulate it?

Read **input**, follow written decisions and place a token in a tray. Record the **trace**. Human **simulation** does not establish model **training**.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Build a runnable paper prototype with explicit instructions and visible state	Explain the deciding step, show a trace or test, and state one limit.
Record what actually happens without silently repairing the instructions	Explain the deciding step, show a trace or test, and state one limit.
Use a trace to distinguish an instruction defect from an execution mistake	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional team turns a storage-bin specification into movable **instruction** cards.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional team turns a storage-bin specification into movable **instruction** cards.
- Which labelled step matches this detail? Expected Y, actual X: locate the missing check.
- 1. Actual **trace**
- 2. Record mismatch
- 3. Operate literally
- 4. **Expected result**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Record mismatch Expected Y, actual X: locate the missing check.

A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Changed-case question

- Changed-case practice: Insert “if capacity is below count, continue” before accepting a bin. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The revised **trace** rejects X and accepts Y for 4.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: The revised **trace** rejects X and accepts Y for 4. 3<4 is true; 7<4 is false, so Y fits.

Repair and save your work

Review Design the Decision **Process** (ai-1/31). Goal: Design a decision **process** that covers valid, invalid and no-match requests.

Short practice: Where does the first original rejection occur, and why is K3 not originally selected?. Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Prompts Are Design **Instruction**s (ai-1/26). Goal: Write a prompt with a clear task, supplied **evidence** and checkable **output** requirements.

Short practice: What must happen if B instead conflicts with another equally authoritative duration record?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

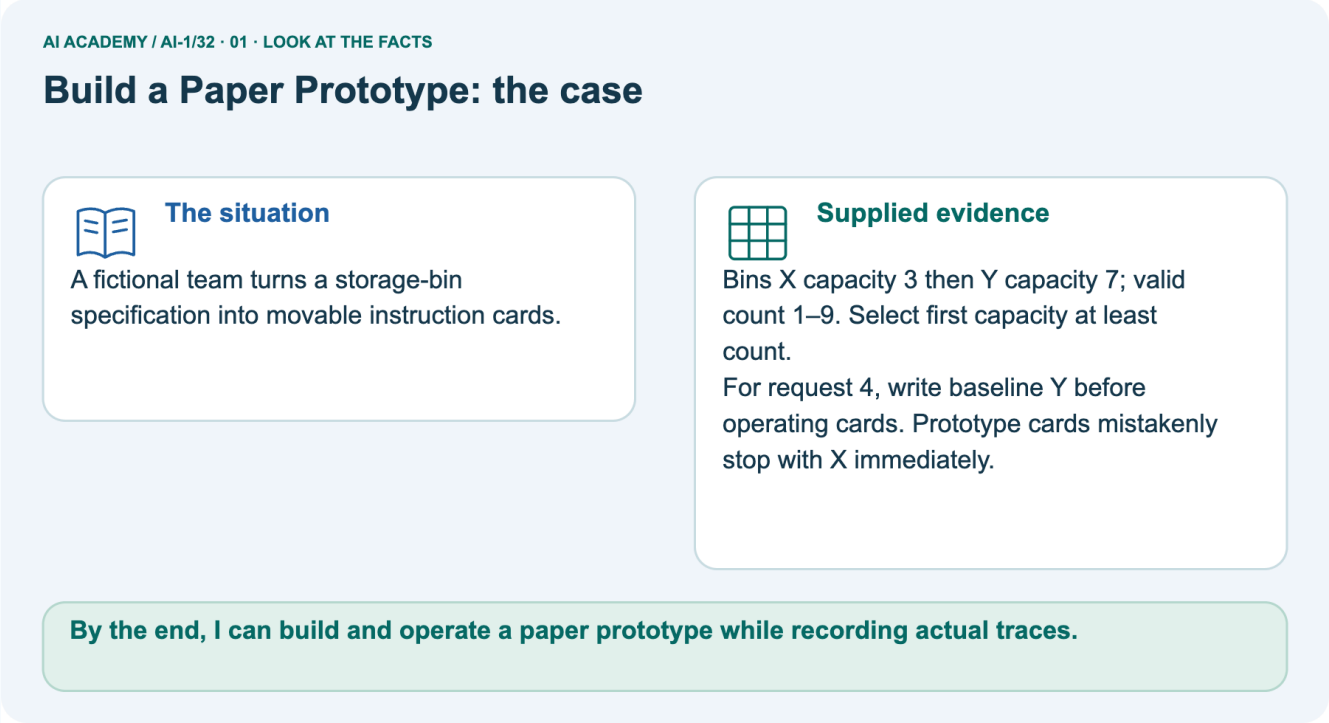
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Build a Paper Prototype: the case



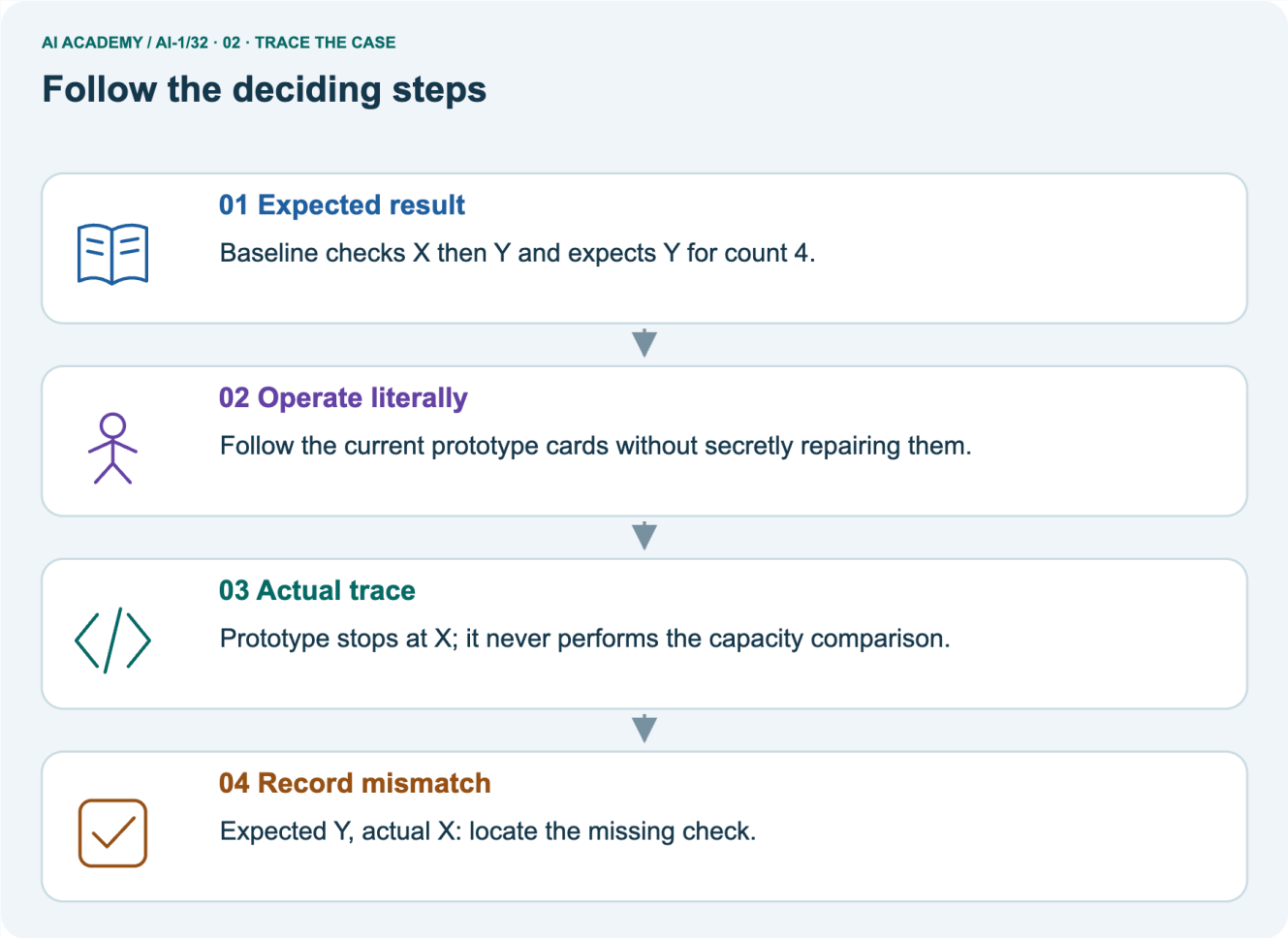
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional team turns a storage-bin specification into movable **instruction** cards. Supplied facts: Bins X capacity 3 then Y capacity 7; valid count 1–9. Select first capacity at least count.

For request 4, write baseline Y before operating cards. **Prototype** cards mistakenly stop with X immediately.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

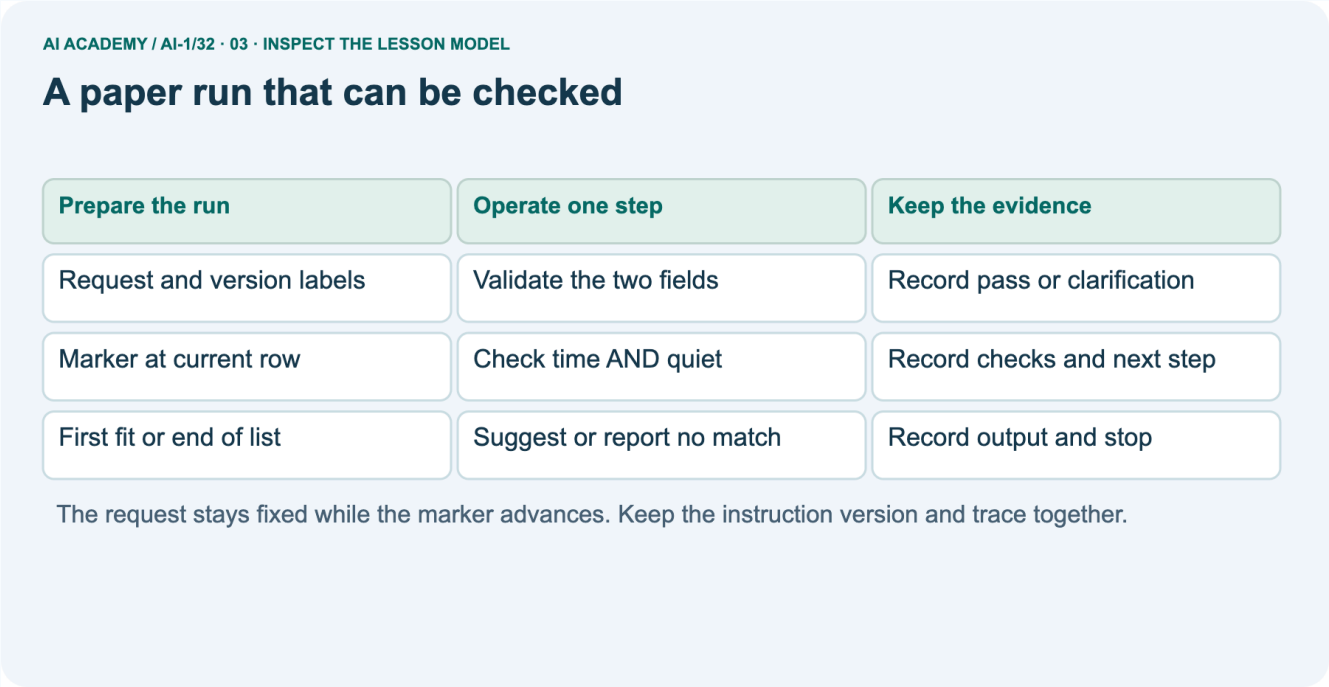


Follow the labelled picture with the matching explanation. Open the vector diagram

Expected result: Baseline checks X then Y and expects Y for count 4. → Operate literally: Follow the current **prototype** cards without secretly repairing them. → Actual **trace**: **Prototype** stops at X; it never performs the capacity comparison. → Record mismatch: Expected Y, actual X: locate the missing check.

Point to each numbered stage and explain which recorded **input** it uses.

03. A paper run that can be checked



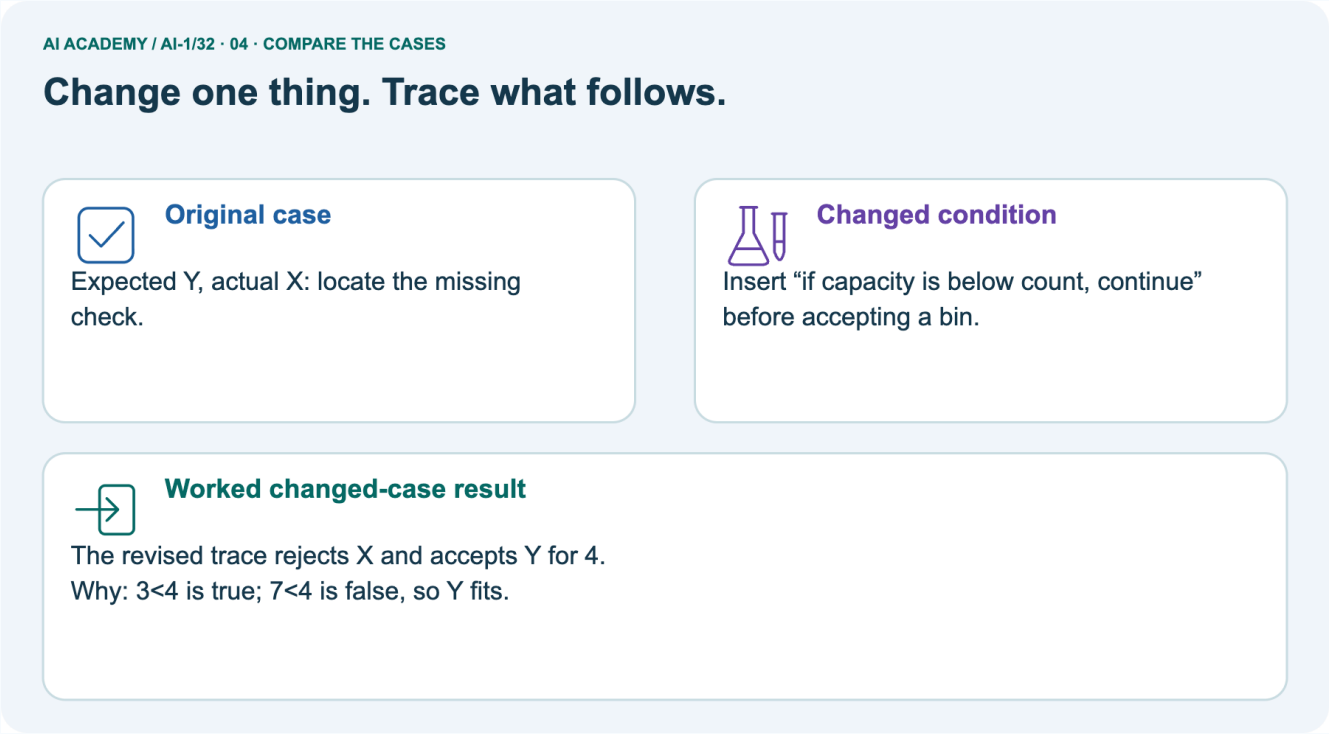
Follow the labelled picture with the matching explanation. Open the vector diagram

A paper run that can be checked. Request and version labels | Validate the two fields | Record pass or clarification. Marker at current row | Check time AND quiet | Record checks and next step.

First fit or end of list | Suggest or report no match | Record **output** and stop

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



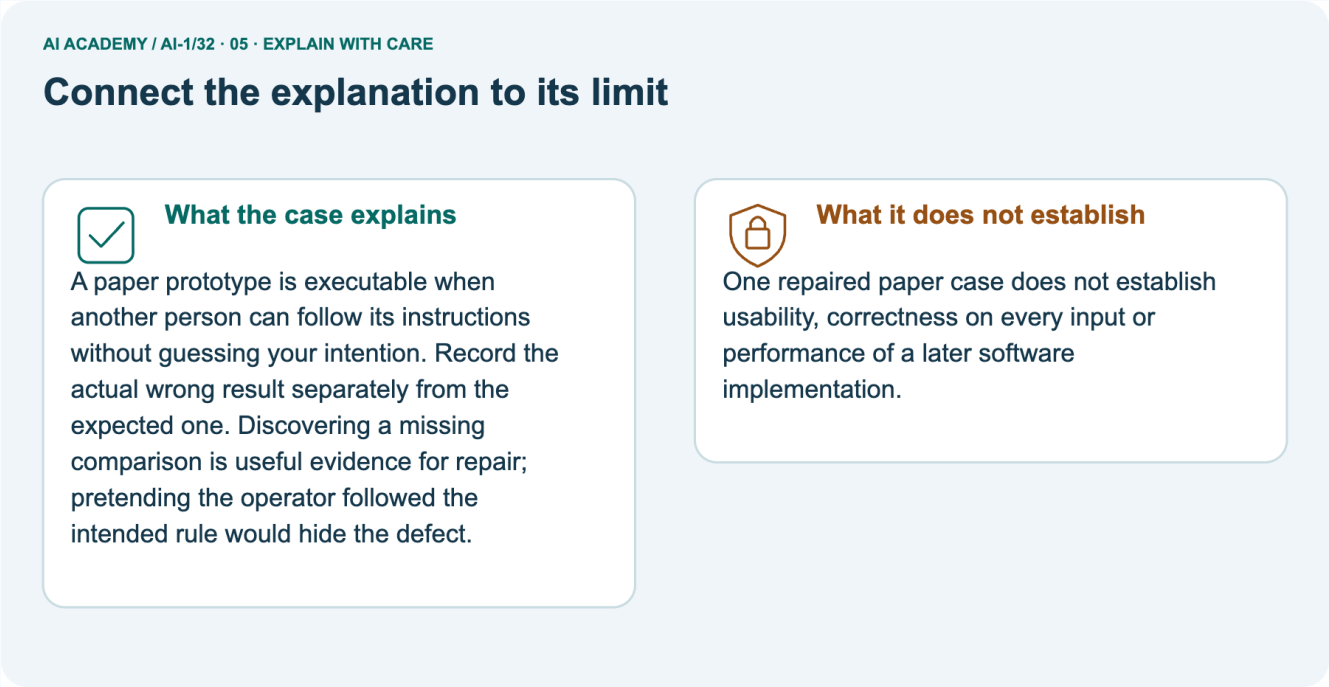
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Expected Y, actual X: locate the missing check. Changed condition: Insert “if capacity is below count, continue” before accepting a bin. **Expected result**: The revised **trace** rejects X and accepts Y for 4.

Reason: 3<4 is true; 7<4 is false, so Y fits.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

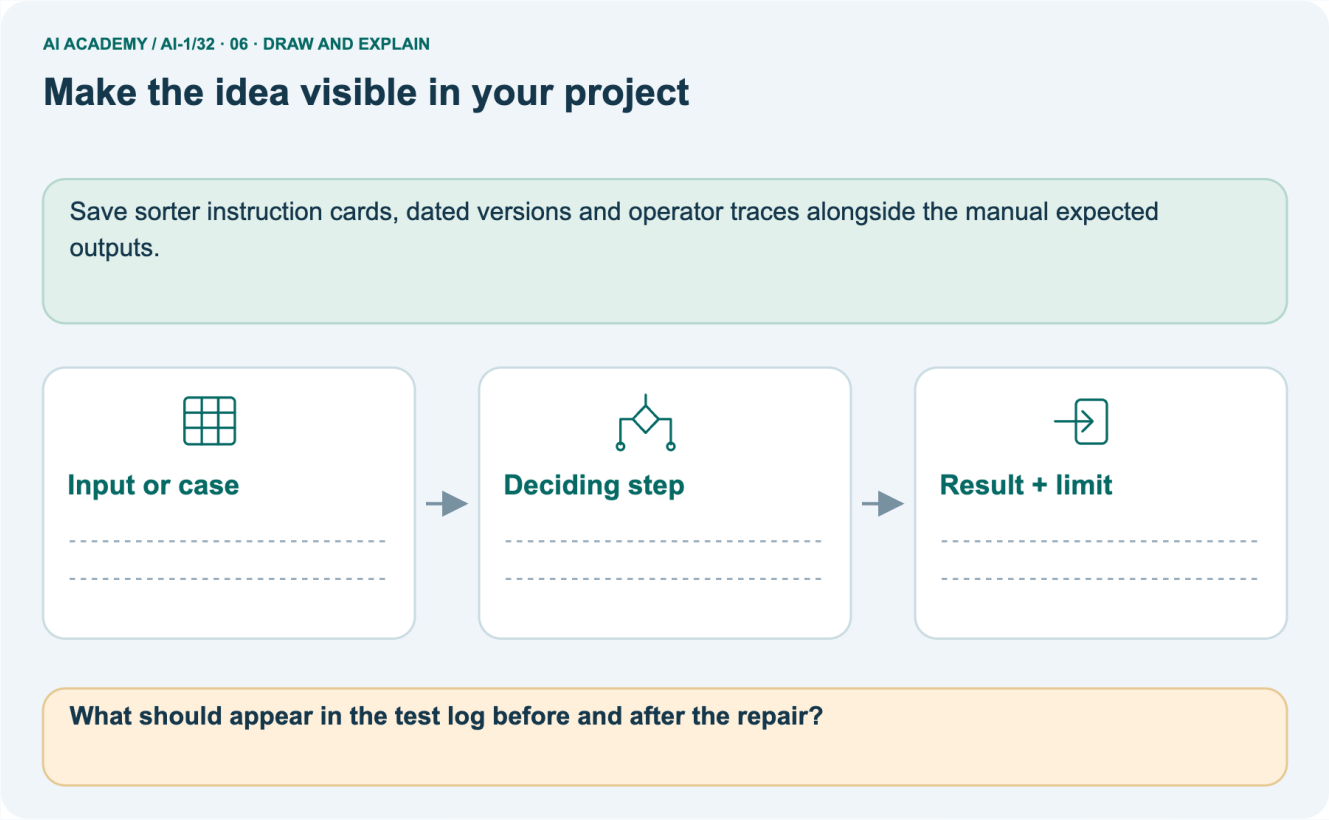
A paper **prototype** is executable when another person can follow its **instruction**s without guessing your intention. Record the actual wrong result separately from the expected one.

Discovering a missing comparison is useful **evidence** for repair; pretending the **operator** followed the intended rule would hide the defect.

Limit: One repaired paper case does not establish usability, correctness on every **input** or performance of a later software implementation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **instruction** cards, dated versions and **operator trace**s alongside the manual expected **output**s. Draw your **input**, deciding step, result and limit.

Supported question: What should appear in the test log before and after the repair?

Draw your project **trace** and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Build a Paper Prototype: the case

02. Follow the deciding steps

03. A paper run that can be checked

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.